

Student Name _____

Spring 2025. Physics 510.

Quantum Theory II.

Test # 2.

Show all your work step by step and write down your final answers explicitly. Plain final answers will not be graded. You are allowed to use your equation sheet. A4, one side, handwritten by you personally.

1. Consider a three-dimensional repulsive ($V_0 > 0$) square barrier of radius a .

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

hyperbolic $\sinh$ (solution)
not
apply B.C. - $\sin$
 $\cos$

A particle of energy $E = \frac{\hbar^2 k^2}{2m} < V_0$ is incident on the well.

- Write down the Schrödinger equation for an s-wave ($l = 0$ mode).
- Solve it directly for both regions (it is possible because $l = 0$). Do not forget that the asymptotically free solution must contain a partial phase shift.
- Derive the phase shift, δ_0 , for the mode by applying necessary boundary conditions.
- How does the phase shift behave if $V_0 \rightarrow \infty$?
- Derive the total cross-section of the mode, σ_0 , in the limit of zero energy.

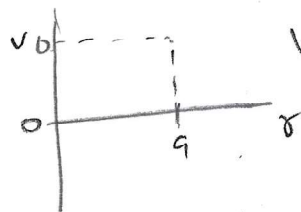
2. Calculate the scattering amplitude and differential cross-section in the Born approximation for the same potential

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

integrate by part.

(1)

consider a 3-d Krulstide ($V_0 > 0$) square barrier of width a ,



$$V(x) = \begin{cases} V_0, & x < a \\ 0, & x \geq a \end{cases}$$

the solution to the Schrodinger's equation for the square barrier becomes, inside the square,

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(x) = E \psi(x)$$

$$k^2 = \frac{2mE}{\hbar^2}$$

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

Applying BC,

$$\psi(0) = 0$$

$$\psi(0) = A \sin(0) + B \cos(0) = 0$$

$$\psi(0) = B = 0$$

$$\psi(x) = A \sin(kx)$$

when $x = a$

Apply BC $\psi(a) = 0$

$$\psi(a) = A \sin(ka) = 0$$

where $k = \frac{n\pi}{a}$ and $n = 1, 2, 3, \dots$ since $ka = n\pi$

$$\psi(x) = A \sin\left(\frac{n\pi}{a}x\right)$$

$$\psi(x) = A \sin\left(\frac{n\pi x}{a}\right)$$

$$\text{Since } k^2 = \frac{2mE}{\hbar^2}$$

$$\frac{n^2\pi^2}{a^2} = \frac{2mE}{\hbar^2}$$

$$E = \frac{n^2\pi^2\hbar^2}{2ma^2}$$

$$\psi_1'(x) = -A \cos\left(\frac{n\pi x}{a}\right)$$

(2)

Outside the square, the solution to the Schrödinger's equation becomes,

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(x) + V(x) \psi(x) = E \psi(x)$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(x) = E \psi(x) - V(x) \psi(x)$$

$\Downarrow$
 V_0

where $k^2 = \frac{2m(E - V_0)}{\hbar^2}$

the solution to the Schrödinger's equation becomes,

$$\psi_2(x) = C \sinh(kx) + B \cosh(kx)$$

Applying bc, $\psi(0) = 0$

$$\psi_2(0) = C \sinh(0) + B \cosh(0) = 0$$

$$\psi_2(0) = B = 0$$

$$\psi_2(x) = C \sinh(kx)$$

when $\psi_2(a) = 0$

$$\psi_2(a) = C \sinh(ka)$$

where $k = \frac{n\pi}{a}$, $n = 1, 2, 3$

$$\psi_2(x) = C \sinh\left(\frac{n\pi x}{a}\right)$$

$$\psi_2'(x) = C \cosh\left(\frac{n\pi x}{a}\right)$$

$$\text{Since } k^2 = \frac{2m(E - V_0)}{\hbar^2}$$

(3)

$$\left(\frac{n\pi}{a}\right)^2 = \frac{2m(E - V_0)}{\hbar^2}$$

$$\frac{n^2 \pi^2}{a^2} = \frac{2m(E - V_0)}{\hbar^2}$$

$$2m a^2 (E - V_0) = \hbar^2 n^2 \pi^2$$

$$2m a^2 E - 2m a^2 V_0 = \hbar^2 n^2 \pi^2$$

$$E = \frac{\hbar^2 n^2 \pi^2 + 2m a^2 V_0}{2m a^2}$$

$$E = \frac{\hbar^2 n^2 \pi^2}{2m a^2} + V_0 \quad \approx \quad E = \frac{\hbar^2 k^2}{2m} < V_0$$

So for $l=0$,

$$V_l = V_0$$

so from the 2nd schrodinger equation,

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(r) + V_0 \psi(r) = E \psi(r)$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi_l(r) + \left[V_0 + \frac{\hbar^2}{2m} \left(\frac{l(l+1)}{r^2} \right) \right] \psi_l(r) = E \psi_l(r)$$

Q when $l=0$

the solution becomes,

$$-\frac{\hbar^2}{2m} \nabla^2 V_0(r) + \left[V_0 + \frac{\hbar^2}{2m} \right] V_0(r) = E V_0(r)$$

(this is for the region outside the square barrier)

(b) for the region inside

$$-\frac{\hbar^2}{2m} \nabla^2 V_0(r) = E V_0(r)$$

⑦ For the phase shift δ_0 , is the difference between the actual asymptotic phase and the free particle phase,

$$V_0(r) \xrightarrow{r \rightarrow \infty} A \sin\left(kr - \frac{\pi}{2} l\right) \quad \begin{array}{l} l \approx 0 \text{ since } l=0 \\ \downarrow \\ \text{angular} \\ \text{momentum} \\ \text{parameter.} \end{array}$$

$$V_0(r) \xrightarrow{r \rightarrow \infty} A \sin(kr)$$

$$V_0^{(0)}(r) \xrightarrow{r \rightarrow \infty} A \sin\left(kr - \frac{\pi}{4}\right) \quad (\text{free particle}).$$

$$\delta_0 = V_0(r) - V_0^{(0)}(r)$$

$$= A \sin(kr) - A \sin\left(kr - \frac{\pi}{4}\right)$$

$$\delta_0 = A \sin \frac{\pi}{4}$$

④ as $V_0 \rightarrow \infty$, the phase shift behaves asymptotically.

⑤

(7)

Problem 2

Scattering amplitude,

$$f(\theta) = f(q) = \frac{m}{2\pi\hbar^2} \int e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r}) d^3\mathbf{r}.$$

where

$$V(r) = \begin{cases} V_0 & \text{for } r < a \\ 0 & \text{for } r \geq a \end{cases}$$

The equation disappears at $V(r) = 0$, so we solve for $V(r) = V_0$

$$f(\theta) = f(q) = \frac{m}{2\pi\hbar^2} \int e^{-i\mathbf{q}\cdot\mathbf{r}} V_0 d^3\mathbf{r}$$

where $q r = q r \cos \theta$

$$d^3\mathbf{r} = r^2 dr d\Omega = r^2 dr \sin \theta d\theta d\phi$$

$$f(\theta) = \frac{m}{2\pi\hbar^2} \int e^{-iqr \cos \theta} \cdot V_0 \cdot r^2 dr \cdot \sin \theta d\theta d\phi$$

$$f(\theta) = \frac{m}{2\pi\hbar^2} \int_0^\pi e^{-iqr \cos \theta} \sin \theta d\theta \cdot \int_0^{2\pi} d\phi \cdot \int_0^a r^2 V_0 dr$$

$\Downarrow$
 polar
 coordinate,
 $\int_2$

$\Downarrow$
 angular
 coordinate
 $\int_1$

10